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Forces and Friction

Draw every force, then resolve in two directions at right angles. Friction is at most μR , and equals μR when limiting.

THE TOPIC ON ONE MAP

Forces and Friction

01 — Equilibrium of a particle

resolve twice

two strings

constant speed

02 — Friction on a horizontal surface

$F \leq \mu R$

pulled at an angle

does it move?

03 — Rings and beads

horizontal rod

vertical wire

which way is friction?

EQUILIBRIUM = RESULTANT FORCE ZERO

→ forces = ← forces

↑ forces = ↓ forces

“constant speed” and “at rest” are both equilibrium
limiting

friction: $F \leq \mu R$, $= \mu R$ when

01 Equilibrium of a Particle: Resolve Twice

Draw the particle as a dot with every force as an arrow. Choose two directions at right angles (usually horizontal and vertical) and write “forces one way = forces the other way” for each. Two equations, two unknowns.

WHEN YOU SEE → YOU DO

“in equilibrium”, “at rest”, “constant speed”

→ resultant force = 0: resolve twice

a force at θ to the horizontal

→ $F \cos \theta$ across, $F \sin \theta$ up

“ $\tan \theta = \frac{3}{4}$ ”

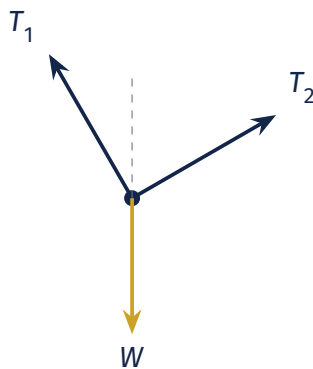
→ $\sin \theta = \frac{3}{5}$, $\cos \theta = \frac{4}{5}$ — exact

forces as vectors $a\mathbf{i} + b\mathbf{j}$

→ add the vectors, total = 0

“smooth ring on a string”

→ the tension is the same on both sides



$$\rightarrow: T_2 \cos 30^\circ = T_1 \cos 60^\circ$$

$$\uparrow: T_1 \sin 60^\circ + T_2 \sin 30^\circ = W$$

WORKED MODEL • JANUARY 2022

P (weight 5 N) on a string at 60° to the vertical, held by F perpendicular to the string. Find F and T .

1 resolve along the string: $T = 5 \cos 60^\circ$

perpendicular pair of directions

2 resolve along F : $F = 5 \sin 60^\circ$

ANSWER $F = \frac{5\sqrt{3}}{2} \approx 4.3 \text{ N}, \quad T = 2.5 \text{ N}$

IDEA 01

Practise: Equilibrium of a Particle: Resolve Twice

My questions first, then real past papers.

Q1 DR ESLAM Two strings

M1 • TOPIC 6 4 marks

A particle of weight 20 N hangs in equilibrium from two strings that make 30° and 60° with the vertical. Find the two tensions.

Resolve horizontally and vertically: two equations, two tensions.

Q2 DR ESLAM Vector forces

M1 • TOPIC 6 2 marks

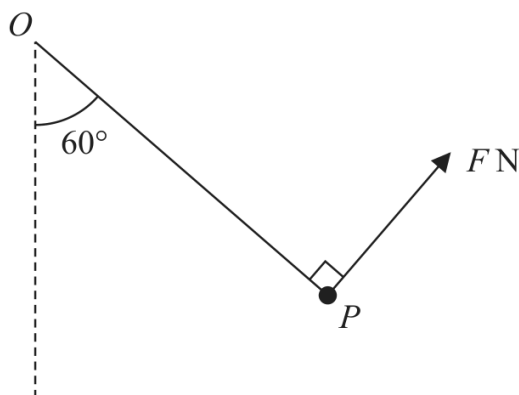
Forces $(4\mathbf{i}+3\mathbf{j})$ N, $(-2\mathbf{i}+5\mathbf{j})$ N and $\mathbf{F}$ are in equilibrium. Find $\mathbf{F}$ and its magnitude.

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 • January 2022 • Q1

M1 • TOPIC 6 6 marks

1.

**Figure 1**

A particle P of weight 5 N is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point O . The particle P is held in equilibrium by a force of magnitude F newtons. The direction of this force is perpendicular to the string and OP makes an angle of 60° with the vertical, as shown in Figure 1.

Find

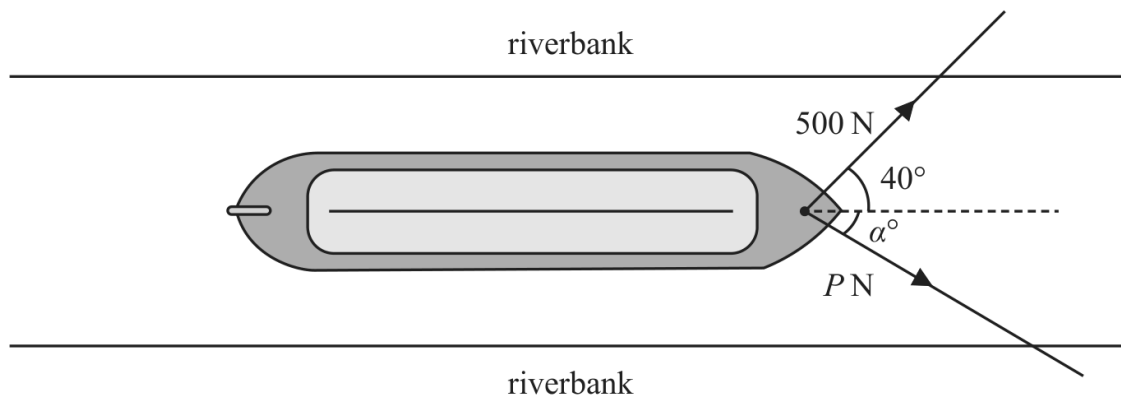
(a) the value of F (3)

(b) the tension in the string. (3)

Q4 PAST PAPER Pearson IAL WME01 • January 2023 • Q6

M1 • TOPIC 6 8 marks

6.

**Figure 3**

A boat is pulled along a river at a constant speed by two ropes.

The banks of the river are parallel and the boat travels horizontally in a straight line, parallel to the riverbanks.

- The tension in the first rope is 500 N acting at an angle of 40° to the direction of motion, as shown in Figure 3.
- The tension in the second rope is P newtons, acting at an angle of α° to the direction of motion, also shown in Figure 3.
- The resistance to motion of the boat as it moves through the water is a constant force of magnitude 900 N

The boat is modelled as a particle. The ropes are modelled as being light and lying in a horizontal plane.

Use the model to find

- the value of α
- the value of P

(8)

Q5 PAST PAPER Pearson IAL WME01 · January 2024 · Q1

M1 · TOPIC 6 6 marks

1.

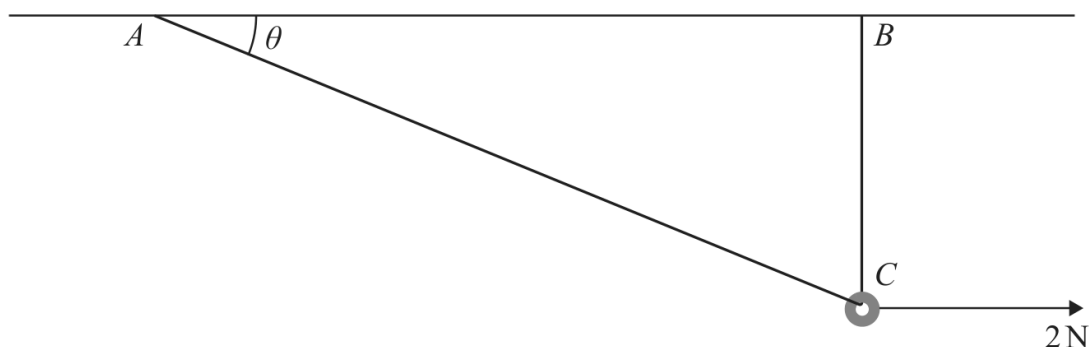
**Figure 1**

Figure 1 shows a small smooth ring **threaded** onto a light inextensible string.

One end of the string is attached to a fixed point A on a horizontal ceiling and the other end of the string is attached to a fixed point B on the ceiling.

A horizontal force of magnitude 2 N acts on the ring so that the ring rests in equilibrium at a point C , vertically below B , with the string taut.

The line of action of the horizontal force and the string both lie in the same vertical plane.

The angle that the string makes with the ceiling at A is θ , where $\tan \theta = \frac{3}{4}$

The tension in the string is T newtons. The mass of the ring is M kg.

(a) Find the value of T (3)

(b) Find the value of M (3)

02 Friction on a Horizontal Surface

First find R by resolving vertically — a pull at an angle makes R smaller, a push makes it bigger. Then friction: at rest, F is whatever is needed (at most μR); on the point of sliding, or moving, $F = \mu R$ exactly.

WHEN YOU SEE → YOU DO

“on the point of sliding” / “limiting equilibrium”

$$\rightarrow F = \mu R$$

“find the friction and describe what happens”

→ find the friction needed, compare with μR

a pull upwards at 30°

$$\rightarrow R = mg - P \sin 30^\circ$$

a push downwards at α

$$\rightarrow R = mg + P \sin \alpha$$

“show that $P \leq \dots$ ”

→ needed friction $\leq \mu R$, rearrange

moving: “find the acceleration”

$$\rightarrow F = \mu R, \text{ then } F = ma$$

Resolve vertically: R

Friction needed

Compare with μR

Conclude

WORKED MODEL • JUNE 2022

5 kg block, $\mu = \frac{3}{7}$, $P = 14$ N at 30° above the horizontal. Does it move?

1 $R = 5g - 14 \sin 30^\circ = 42$

resolve vertically

2 needed: $14 \cos 30^\circ = 12.1$; available: $\frac{3}{7}(42) = 18$

compare

ANSWER $F = 12.1$ N; the block stays at rest

× **TRAP** Writing $R = mg$ when the force is at an angle. Always resolve vertically first.

IDEA 02

Practise: Friction on a Horizontal Surface

My questions first, then real past papers.

Q6 DR ESLAM Does it move?

M1 • TOPIC 6 3 marks

A 10 kg box rests on a rough horizontal floor, $\mu = 0.4$. A horizontal force of 30 N acts on it. Does it move? Find the friction.

Compare the push with μR (the most friction can give).

Q7 DR ESLAM Pulled at an angle

M1 • TOPIC 6 4 marks

A 5 kg box, $\mu = 0.5$, is pulled by a force P at 30° above the horizontal and is on the point of sliding. Find P .

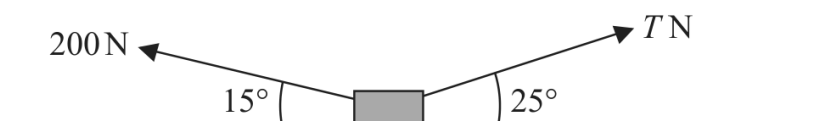
The pull lifts the box a little: $R = 5g - P \sin 30^\circ$.

PAST PAPERS

Q8 PAST PAPER Pearson IAL WME01 · January 2021 · Q3

M1 · TOPIC 6 9 marks

3.

**Figure 1**

A parcel of mass 20 kg is at rest on a rough horizontal floor. The coefficient of friction between the parcel and the floor is 0.3

Two forces, both acting in the same vertical plane, of magnitudes 200 N and T N are applied to the parcel. The line of action of the 200 N force makes an angle of 15° with the horizontal and the line of action of the T N force makes an angle of 25° with the horizontal, as shown in Figure 1. The parcel is modelled as a particle P .

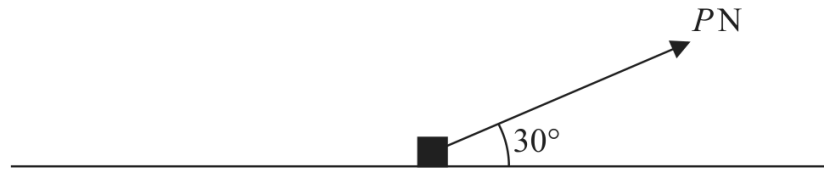
Find the smallest value of T for which P remains in equilibrium.

(9)

Q9 PAST PAPER Pearson IAL WME01 · June 2022 · Q4

M1 · TOPIC 6 12 marks

4.

**Figure 1**

A small block of mass 5 kg lies at rest on a rough horizontal plane.

The coefficient of friction between the block and the plane is $\frac{3}{7}$

A force of magnitude P newtons is applied to the block in a direction which makes an angle of 30° with the plane, as shown in Figure 1.

The block is modelled as a particle.

Given that $P = 14$

- (a) find the magnitude of the frictional force exerted on the block by the plane and describe what happens to the block, justifying your answer. (6)

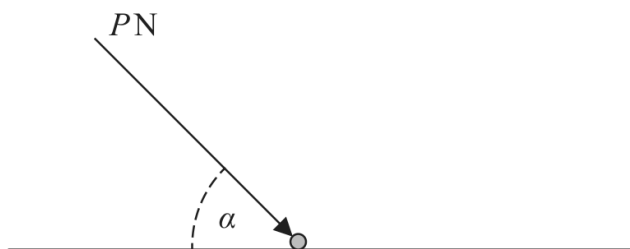
The value of P is now changed so that the block is on the point of slipping along the plane.

- (b) Find the value of P (6)

Q10 PAST PAPER Pearson IAL WME01 • June 2023 • Q6

M1 • TOPIC 6 7 marks

6.

**Figure 3**

A particle of weight W newtons lies at rest on a rough horizontal surface, as shown in Figure 3.

A force of magnitude P newtons is applied to the particle.

The force acts at an angle α to the horizontal, where $\tan \alpha = \frac{4}{3}$

The coefficient of friction between the particle and the surface is $\frac{1}{4}$

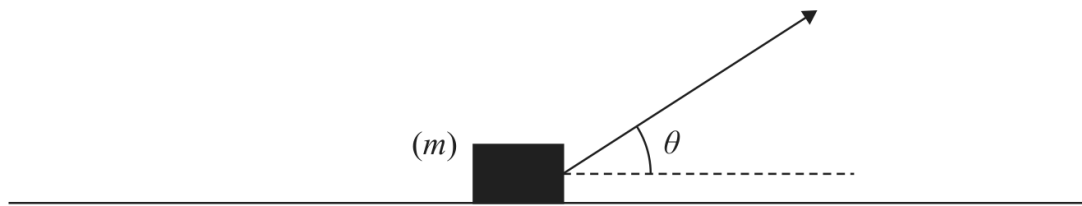
Given that the particle does not move, show that

$$P \leq \frac{5W}{8} \quad (7)$$

Q11 PAST PAPER Pearson IAL WME01 · June 2024 · Q6

M1 · TOPIC 6 12 marks

6.

**Figure 3**

A box of mass m lies on a rough horizontal plane. The box is pulled along the plane in a straight line at **constant speed** by a light rope. The rope is inclined at an angle θ to the plane, as shown in Figure 3.

The coefficient of friction between the box and the plane is $\frac{1}{3}$

The box is modelled as a particle.

Given that $\tan \theta = \frac{3}{4}$

(a) find, in terms of m and g , the tension in the rope.

(7)

The rope is now removed and the box is placed at rest on the plane. The box is then projected horizontally along the plane with speed u .

The box is again modelled as a particle.

When the box has moved a distance d along the plane, the speed of the box is $\frac{1}{2}u$.

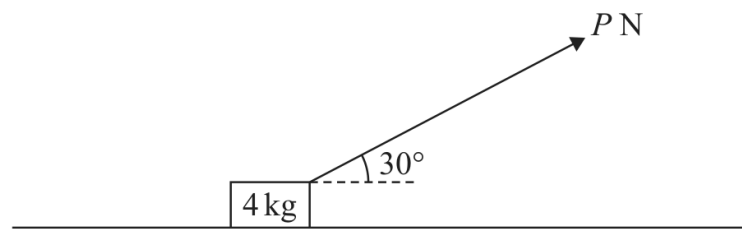
(b) Find d in terms of u and g .

(5)

Q12 PAST PAPER Pearson IAL WME01 • October 2025 • Q5

M1 • TOPIC 6 13 marks

5.

**Figure 3**

A box of mass 4 kg is placed on a rough horizontal surface.

A force of magnitude P newtons, acting at 30° to the horizontal, is applied to the box, as shown in Figure 3.

The coefficient of friction between the box and the surface is $\frac{2}{3}$

The box is modelled as a particle.

(a) Find the value of P when the box is on the point of sliding along the surface. (6)

The value of P is now increased to 25 and the box moves along the surface.

Find

(b) the acceleration of the box, (5)

(c) the speed of the box when it has moved 1.5 m. (2)

03 Rings and Beads on Rods and Wires

The normal reaction is **perpendicular to the rod**. On a horizontal rod it is vertical; on a vertical wire it is horizontal. Friction acts along the rod, against the way the ring would slide.

WHEN YOU SEE → YOU DO

“threaded on a rough horizontal rod/wire”

→ R vertical, friction horizontal

“threaded on a rough vertical rod/wire”

→ R horizontal, friction vertical

“state the direction of the friction”

→ opposite to the way it would move without friction

“the tension can vary — min and max”

→ about to slip down: friction up; about to slip up: friction down

two rings joined by a string with a mass below

→ resolve for the mass first, then each ring

WORKED MODEL • OCTOBER 2023

Ring 0.2 kg on a rough vertical rod; string at θ to the rod, $\tan \theta = \frac{12}{5}$, tension 10 N. Friction?

1 up the rod: $10 \cos \theta = 10 \cdot \frac{5}{13} = 3.85$

tension's vertical part

2 down: weight $0.2g = 1.96$

ANSWER $F = 3.85 - 1.96 \approx 1.89$ N, acting downwards

IDEA 03

Practise: Rings and Beads on Rods and Wires

My questions first, then real past papers.

Q13 DR ESLAM Ring on a horizontal rod

M1 · TOPIC 6 4 marks

A ring of mass 0.5 kg on a rough horizontal rod ($\mu = 0.2$) is pulled by a string at 40° above the rod and is about to slide. Find the tension.

Q14 DR ESLAM Bead on a vertical wire

M1 · TOPIC 6 3 marks

A bead of weight 3 N on a rough vertical wire is held at rest by a horizontal force of 12 N. Find the least possible value of μ .

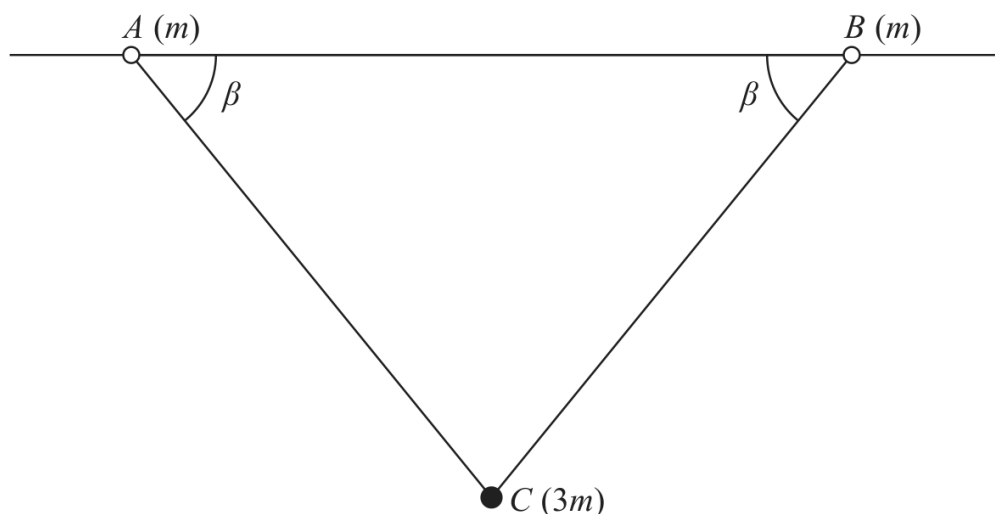
On a vertical wire the normal reaction is horizontal.

PAST PAPERS

Q15 PAST PAPER Pearson IAL WME01 • January 2020 • Q4

M1 • TOPIC 6 10 marks

4.

**Figure 2**

Two identical small rings, A and B , each of mass m , are threaded onto a rough horizontal wire. The rings are connected by a light inextensible string. A particle C of mass $3m$ is attached to the midpoint of the string. The particle C hangs in equilibrium below the wire with angle $BAC = \beta$, as shown in Figure 2.

The tension in each of the parts, AC and BC , of the string is T

(a) By considering particle C , find T in terms of m , g and β (2)

(b) Find, in terms of m and g , the magnitude of the normal reaction between the wire and A . (3)

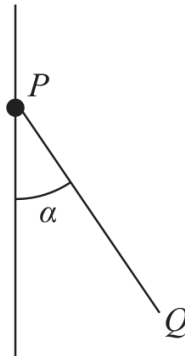
The coefficient of friction between each ring and the wire is $\frac{4}{5}$

The two rings, A and B , are on the point of sliding along the wire towards each other.

(c) Find the value of $\tan \beta$ (5)

Q16 PAST PAPER Pearson IAL WME01 • October 2021 • Q5**M1 • TOPIC 6 10 marks**

5.

**Figure 2**

A small bead of mass 0.2 kg is attached to the end P of a light rod PQ . The bead is threaded onto a fixed vertical rough wire.

The bead is held in equilibrium with the rod PQ inclined to the wire at an angle α , where $\tan \alpha = \frac{4}{3}$, as shown in Figure 2.

The thrust in the rod is T newtons.

The bead is modelled as a particle.

- (a) Find the magnitude and direction of the friction force acting on the bead when $T = 2.5$
- (3)**

The coefficient of friction between the bead and the wire is μ .

Given that the greatest possible value of T is 6.125

- (b) find the value of μ .
- (7)**

5.

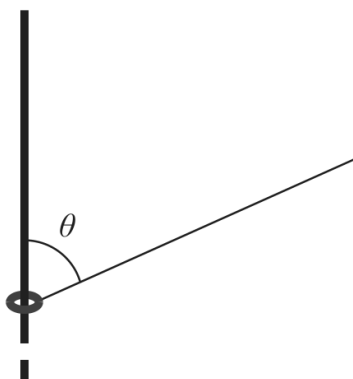


Figure 3

A small ring of mass 0.2 kg is attached to one end of a light inextensible string.

The ring is **threaded** onto a fixed rough vertical rod.

The string is taut and makes an angle θ with the rod, as shown in Figure 3,
where $\tan \theta = \frac{12}{5}$

Given that the ring is in equilibrium and that the tension in the string is 10 N ,

(a) find the magnitude of the frictional force acting on the ring, (3)

(b) state the direction of the frictional force acting on the ring. (1)

The coefficient of friction between the ring and the rod is $\frac{1}{4}$

Given that the ring is in equilibrium, and that the tension in the string, T newtons, can now vary,

(c) (i) find the minimum value of T
(ii) find the maximum value of T (8)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|--|
| 01 | Equilibrium | Resolve in two directions; two equations. |
| 02 | Friction | R first; $F \leq \mu R$; $= \mu R$ when limiting or moving. |
| 03 | Rings and beads | R perpendicular to the rod; friction against the slide. |

BOTTOM LINE Diagram $\rightarrow$ resolve $\rightarrow R \rightarrow F \leq \mu R$. Limiting means $F = \mu R$.

Answers

Forces and Friction • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • EQUILIBRIUM OF A PARTICLE: RESOLVE TWICE

- Q1** Dr Eslam • Two strings $T_{30} = 10\sqrt{3} \approx 17.3 \text{ N}, T_{60} = 10 \text{ N}$
- Q2** Dr Eslam • Vector forces $\mathbf{F} = -2\mathbf{i} - 8\mathbf{j}; \sqrt{68} \approx 8.25 \text{ N}$
- Q3** Pearson IAL WME01 • January 2022 • Q1 (a) $F = \frac{5\sqrt{3}}{2} \approx 4.3 \text{ N}$ (b) $T = 2.5 \text{ N}$
- Q4** Pearson IAL WME01 • January 2023 • Q6 (i) $\alpha \approx 31.9^\circ$ (ii) $P \approx 609 \text{ N}$
- Q5** Pearson IAL WME01 • January 2024 • Q1 (a) $T = 2.5 \text{ N}$ (b) $M \approx 0.408$

IDEA 02 • FRICTION ON A HORIZONTAL SURFACE

- Q6** Dr Eslam • Does it move? $\mu R = 39.2 > 30$: it stays; $F = 30 \text{ N}$
- Q7** Dr Eslam • Pulled at an angle $P \cos 30^\circ = 0.5(49 - 0.5P) \Rightarrow P \approx 22.0 \text{ N}$
- Q8** Pearson IAL WME01 • January 2021 • Q3 $T = 192 \text{ N}$
- Q9** Pearson IAL WME01 • June 2022 • Q4 (a) $F = 12.1 \text{ N}$ (less than μR): the block stays at rest (b) $P \approx 19.4$
- Q10** Pearson IAL WME01 • June 2023 • Q6 $R = W + \frac{4}{5}P$, $\frac{3}{5}P \leq \frac{1}{4}(W + \frac{4}{5}P) \Rightarrow P \leq \frac{5W}{8}$
- Q11** Pearson IAL WME01 • June 2024 • Q6 (a) $T = \frac{mg}{3}$ (b) $d = \frac{9u^2}{8g}$
- Q12** Pearson IAL WME01 • October 2025 • Q5 (a) $P \approx 21.8$ (b) 0.963 m s^{-2} (c) 1.70 m s^{-1}

IDEA 03 • RINGS AND BEADS ON RODS AND WIRES

- Q13** Dr Eslam • Ring on a horizontal rod $T \cos 40^\circ = 0.2(4.9 - T \sin 40^\circ) \Rightarrow T \approx 1.10 \text{ N}$
- Q14** Dr Eslam • Bead on a vertical wire $R = 12$; $3 \leq 12\mu \Rightarrow \mu \geq 0.25$
- Q15** Pearson IAL WME01 • January 2020 • Q4 (a) $T = \frac{3mg}{2 \sin \beta}$ (b) $R = \frac{5mg}{2}$ (c) $\tan \beta = \frac{3}{4}$
- Q16** Pearson IAL WME01 • October 2021 • Q5 (a) 0.46 N upwards (b) $\mu = 0.35$
- Q17** Pearson IAL WME01 • October 2023 • Q5 (a) $F \approx 1.89 \text{ N}$ (b) downwards (c)(i) $T_{\min} \approx 3.19 \text{ N}$ (ii) $T_{\max} \approx 12.7 \text{ N}$